



(SSUET/QR/111)

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY
DEPARTMENT
BS Computer Science
COURSE INFORMATION SHEET

Session:	Fall-2025
Course Title:	Human Computer Interaction
Course Code:	CS-475
Credit Hours:	3
Semester:	7 th
Pre-Requisite:	None
Instructor Name:	Ms. Farhana Ashraf
Email and Contact Information:	fashraf@ssuet.edu.pk ; HS-12B (cubical-01)
WhatsApp Group:	None
Office Hours:	Mondays to Fridays 11:30 am to 12:30 pm
Mode of Teaching:	Synchronous/Asynchronous/Hybrid/Blended

COURSE OBJECTIVE:

The objective of this course is to explain the concepts of HCI. Furthermore, to acquire the knowledge and skills needed to create highly usable software systems.

COURSE OUTLINE:

The scope of HCI and Interaction Design, Usability paradigm and principles, Theory and methods of the design of interactions between people and computers, Ergonomics - human aspects of interaction design, Technological aspects of interface design and design methodologies, Psychology of HCI, HCI laws, Cognitive aspects, Ethnographic Techniques, Organization and Visual Composition, Social and Ethical Implications, HCI Design Concepts, Participatory (user-centered) design and prototyping, Contextual Design



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COURSE LEARNING OUTCOMES (CLOs):

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
1	Recognize key aspects of human computer interaction.	PLO_2 (Knowledge for Solving Computing Problems)	C2 (Understanding)
2	Demonstrate an awareness of ethical considerations in technology design.	PLO_8 (Computing Professionalism and Society)	C2 (Understanding)
3	Interpret design ideas through a user centered design approach.	PLO_4 (Design / Development of Solutions)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	7 (70%)	3 (30%)	-----
Assignments	6 (60%)	-----	4 (40%)
Mid	13 (43.33%)	17 (56.67%)	-----
Final	24 (48 %)	-----	26 (52%)
Total	50 (50 %)	20 (20 %)	30 (30 %)

GRADING POLICY:

Assessment Tools	Percentage	Marks
Quizzes	10%	10
Assignments	10%	10
Midterm Exam	30%	30
Final Exam	50%	50
TOTAL	100%	100

BENCHMARK:

- Interaction Design Foundation (2010). UX Design Courses & Global UX Community. [online] The Interaction Design Foundation. Available at: <https://www.interaction-design.org/>.
- Anu.edu.au. (2022). Human-Computer Interaction - ANU. [online] Available at: <https://programsandcourses.anu.edu.au/2022/course/COMP6390#learning-outcomes> [Accessed 27 Sep. 2024].



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Relevant Job/Skill: The course helps to build the basis of core concepts that will be incorporated in their job skills as:

Interaction / User Experience (UX) Designer: User Experience (UX) Designers focus on creating optimal experiences for users interacting with digital products, such as websites, apps, and software. This role is ideal for HCI majors who are passionate about understanding user behaviors and crafting interfaces that enhance usability and accessibility.

Usability Analyst: Usability Analysts play a critical role in the design and improvement of user interfaces and experiences, ensuring that products are intuitive, accessible, and enjoyable. This role is perfect for HCI majors who are passionate about making technology user-friendly and advocating for the end-user's needs.

Supporting Software Tool: Figma.

Text Books:

- Dix, A., Finlay, J., Abowd, G.D. and Beale, R. (2004). Human-computer interaction. Harlow: Pearson Prentice Hall, Cop.
- Sharp, H., Rogers, Y. and Preece, J. (2019). Interaction Design : beyond human-computer Interaction. 5th ed. Indianapolis, In: Wiley.

Reference Books:

- Nielsen, J. (1993). Usability Engineering. Amsterdam: Morgan Kaufmann.
- Norman, D. (2013). The Design of Everyday Things. [online] Massachusetts: Mit Press. Available at:
<https://ia902800.us.archive.org/3/items/thedesignofeverydaythingsbydonnorman/The%20Design%20of%20Everyday%20Things%20by%20Don%20Norman.pdf>.



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LESSON PLAN

Course Title: Human Computer Interaction

Course Code: CS-475

Week No.	Week Dates	Topics	Required Reading	Key Date
1	14-10-2025 to 17-10-2025	Course Outline, Books and Marks distribution Learning Objectives of the course. Introduction to Human Computer Interaction.	Dix- Introduction: Pg. 1-8	
2	20-10-2025 to 24-10-2025	Usability and User Experience Design.	Nie-Chap2	Assignment # 1
3	27-10-2025 to 31-10-2025	Interaction Design., Design Heuristics and Usability Principles	Sha-Chap1	Quiz # 1
4	03-11-2025 to 07-11-2025	Design Heuristics and Usability Principles, The problem space and Conceptual Model.	Sha-Chap1 Nor-Chap1 Nor-Chap2	
5	10-11-2025 to 14-11-2025	Interface Metaphors & Interaction paradigms.	Sha-Chap2	Quiz # 2
6	17-11-2025 to 21-11-2025	Ergonomics		Assignment # 2
7	24-11-2025 to 28-11-2025	Ergonomics, HCI Laws	Dix-Chap 1	
8	Midterm Examination (01-12-2025 to 06-12-2025)			



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9	08-12-2025 to 12-12-2025	Applied Psychology of User	Dix-Chap1 Pg 53-55	
10	15-12-2025 to 19-12-2025	User Acceptance, Universal Usability	Dix-Chap10	
11	22-12-2025 to 26-12-2025	HCI Design Concepts - Participatory (user-centered) design and Contextual Design	Nie-Chap 4	Quiz # 3
12	29-12-2025 to 02-01-2026	User Analysis	Dix-Chap 5 Pg 197-201	
13	05-01-2026 to 09-01-2026	Task Analysis	Dix-Chap 15	
14	12-01-2026 to 16-01-2026	Prototyping	Sha-Chap 8	Assignment 3
15	19-01-2026 to 23-01-2026	Evaluation	Dix-Chap 9	
16	26-01-2026 to 30-01-2026	Ethnographic Techniques	Nie-Chap 7	
17	Final Examination (03-02-2026 to 14-02-2026)			

Dix: Dix, **Nie:** Nielsen; **Nor:** Norman, **Sha:** Sharp



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Complete Analysis of Course and Level of Skills ensured in:

Technology	The course effectively utilizes HCI concepts and industry-relevant tools like 'Figma'
Emerging Development Paradigms	The course is structured to align with modern computing trends by covering essential HCI principles.
Pertaining to Industry	The topics covered interaction design jobs available in industry.
Modeling and Design	The course is well-structured, progressing logically from basic to advanced concepts. It integrates theoretical learning with practical applications.

Review Committee
Member – 1 Name and
Signature:

Date:

Review Committee
Member – 2 Name and
Signature:

Date:

Review Committee
Member – 3 Name and
Signature:

Date:

Course Coordinator's
Name and Signature:

Date:

Chairperson's Name and
Signature:

Date: